

Adding Integro Differential Equations and Improving symbolic representation of geometry in PINNs

GSOC 2021

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Abstract

NeuralPDE.jl is a Julia package that employs neural networks to obtain a solution of high dimensional differential equations and hence overcoming the curse of dimensionality. One of the techniques that uses Scientific Machine Learning to solve PDEs is Physics Informed Neural Networks (PINNs). The project aims at extending the application of the PINN algorithm to integro-differential equations. The project also propose to add support for constrained equations and different geometry for PINNs.

1 Project Description

NeuralPDE.jl employs PINN algorithm for solving partial differential equations. NeuralPDE.jl combined with ModellingToolkit.jl aims towards symbolic representation of PDEs. The project aims to extend this symbolic representation and the solver for integro differential equations. The subproject 2 aims towards adding support for constrained equations. It also aims to add support for different domains, that will improve the geometry of PINNs.

1.1 Subproject 1 : Solving Integro Differential Equations

This will aim towards solving the following type of equations using PINNs.

$$\frac{dy}{dx} + y(x) = \int_0^x e^{t-x} y(t) dt \quad (1)$$

According to [1], we can evaluate the integral using Gaussian Quadrature method as

$$\int_0^x e^{t-x} y(t) dt = \sum_{i=1}^n w_i e^{t_i(x)-x} y(t_i(x)) \quad (2)$$

This will begin with the implementation of an Integral operator with Symbolics.jl. Like the Differential() operator, there is an Integral operator required. The

integral operator will then be parsed using another helper function. In order to approximate the integration, Quadrature.jl can be used. Or even FastGaussQuadrature.jl can also be used. An example using Quadrature.jl to evaluate the above integral should look like

```
function integral_rhs(x)
    f(t,p) = exp(t - x)*y(t) #taken from Integral operator
    prob = QuadratureProblem(f,0,x)
    sol = solve(prob,HCubatureJL(), reltol=1e-3, abstol=1e-3)
    return sol.u
end
```

The x will be replaced with the corresponding variables or parameters that will be obtained from the Integral operator.

1.2 Subproject 2 : Adding support for Constrained Equations and improving geometry.

This subproject shall focus on adding a support for equations with additional constraints. The ConstrainedEquation structure is already present in ModellingToolkit.jl. This can be used as follows :

```
ConstrainedEquation([x > 0, y < 1/2], u(x,y) ~ x + y^2)
```

The first vector argument contains the constraints and the next one is the equation. In order to deal with the constrained equations, the *generate_training_sets* function is needed to handle the constraints. When the data points are sampled for the loss function corresponding to this equation, the constraints will be taken into the account.

This also includes adding support for different domains for PDEs. These include, unit circle, disks, sphere, product domain, L shaped domain example as mentioned in [1].

Currently, these domains exist in DomainSets.jl however in order to sample points from them additional functions are needed.

1.3 Project Mentors

Christopher Rackauckas (@ChrisRackauckas) will be mentoring me for this project during the summers.

2 Timeline

- **April 13 - May 13 : Initial Phase**

In this period I will finish the existing work with PINN regarding documentations and efficient calculation of derivatives.

- **May 17 - June 6 : Community Bonding**

As this is the community bonding period, I will get involved with the community, start to know people better and shall participate in discussions regarding NeuralPDE.jl or even related projects. Meanwhile, I shall also start working

on my first subproject

- **June 7 - July 12 : Coding Phase**

The coding phase begins. This is the part where I will complete the basic implementation of subproject 1. I shall also write tests for the algorithm in this period.

- **July 13 - July 16 : Phase 1 Evaluation**

The subproject would be presented for evaluation.

- **July 16 - August 16 : Work Period**

- I will make the required changes as per the evaluation, and thus complete the subproject 1
- I will start working on subproject 2 and complete the implementation of required functionality during this period

- **August 16 - August 23 : Last Week**

I will finish off the subprojects and entail documentation if required.

- **August 23 : Phase 2 Evaluation**

The project will be submitted for final evaluation.

3 Previous Contributions

- JSoC 2020

I was selected for JSoC 2020, during which I contributed the following

- Implemented a solver for Kolmogorov Equations
- Implemented a solver for Optimal Stopping Problems
- Added upper and lower bounds for the Deep BSDE solver

- Full Kolmogorov Solver

<https://github.com/SciML/NeuralPDE.jl/pull/202>

<https://github.com/SciML/NeuralPDE.jl/pull/266>

- Tracking parameters in PINNs

<https://github.com/SciML/NeuralPDE.jl/pull/256>

4 Deliverables

I plan to submit the results of the above subprojects using test-cases , plots and notebooks. The following will be the deliverables :

- Extending support for integro differential equations. Adding support for constrained equations and different domains
- Writing all the necessary tests for each implementation. These tests include working tests , high dimensional tests and convergence tests.
- Presentation of plots for the solutions of the tests and the convergence plots for the algorithm.
- Addition of all the implemented addition to the documentation.

5 About Me

I am an undergrad at the Indian Institute of Technology, Roorkee. I am an active member of SDS Labs at IIT Roorkee. We are a bunch of passionate development and design enthusiasts trying to foster software development culture in the campus. While working on our projects, we adhere to stringent timelines and make sure to follow the best practices to create impacts of the highest significance. Likewise inspired by Papers we Love by MIT, we hold Papers we Read in IIT Roorkee, where we discuss and implement research papers from different domains such as Machine Learning, Cryptography, and Operating Systems. Due to this, I got introduced to machine learning and topics such as Fuzzy Neural Networks and Capsule Neural Networks.

I am interested in the field of Machine Learning. And thus was introduced to Julia due to Flux.jl . Soon I realized Julia is vast, and there are many exciting things that I could contribute on. I was part of the JSOC 2020. This was a great learning experience and thus I want to work more with the community.

5.1 Academic Details

- University - Indian Institute of Technology, Roorkee
- Major - Electronics and Communication Engineering
- CGPA - 8.6

5.2 Contact

- Email - a.bharambe123@gmail.com
- Github - ashutosh-b-b

6 Summer Logistics

I won't be having any engagements during the summers and thus I will devote my complete time to this project. I will manage to work around 6 hours on a week-day and around 7 hours for a weekend.

References

- [1] Lu Lu et al. “DeepXDE: A Deep Learning Library for Solving Differential Equations”. In: *SIAM Review* 63.1 (2021), pp. 208–228.